

Chapter 1 Fluid Mechanics Identities

1.1 Reynolds number

The Reynolds number is the ratio of inertial to viscous forces.

Eq. 1 $Re = \rho * V * D / \mu$

Assumption: single characteristic velocity and length.

Valid for internal and external viscous flows.

Figure 1: Channel flow schematic.

Table 1: Symbol definitions for Eq. 1.

Bibliographic identity: Bartz is a named public heat-transfer correlation.